A SUFFICIENTLY LARGE BASE FIELD EXTENSION MAKES A SCHEME JACOBSON AND MAKES RESIDUE FIELDS AT CLOSED POINTS ALGEBRAIC

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter text immediately before the lemma explains that such a base field extension is needed when X is not locally of finite type; that remark lies outside the counted span and enters only as context.

Editorial note. The counted spans carry 4 navigation stubs and no bibliography citation.

Editorial note. The preamble is the chapter's retained minimal preamble (sources/stacks-varieties/retained-preamble.tex lines 1-77), a verbatim excerpt of the native preamble.

Editorial note. The label lemma-make-Jacobson is defined only in varieties.tex, so no cross-chapter restatement had to be weighed for this card.

Allowed standard prerequisites (editorial).

- Base change of Jacobson rings along a sufficiently large field extension. Algebra, Lemma algebra-lemma-base-change-Jacobson, invoked at proof lines 2714-2715. Supplies the cardinal threshold for an affine piece.
- Affine criterion for a Jacobson scheme. Properties, Lemma properties-lemma-affine-jacobson, invoked at proof lines 2716-2718. Supplies the two desired properties for an affine piece beyond the threshold.
- Local criterion for a scheme to be Jacobson. Properties, Lemma properties-lemma-locally-jacobson and Definition properties-definition-jacobson, invoked at proof lines 2722-2723 and named in the statement. Supplies that the property can be checked on the members of an affine open covering.

Source reference. Original lemma 046V: sources/stacks-chow/tags.txt, source lines 5486–5486. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 01P2: sources/stacks-chow/tags.txt, source lines 2215–2215. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 01P3: sources/stacks-chow/tags.txt, source lines 2216–2216. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

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Source reference. Original lemma 01P4: sources/stacks-chow/tags.txt, source lines 2217–2217. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let X be a scheme over a field k. For any field extension K/k whose cardinality is large enough we have*

(1) *for any closed point $x \in X_K$ the extension $\kappa(x)/K$ is algebraic, and*
(2) *X_K is a Jacobson scheme (Properties, Definition 01P2).*

Proof. Choose an affine open covering $X = \bigcup U_i$. By Algebra, Lemma 046V and Properties, Lemma 01P3 there exist cardinals κ_i such that $U_{i,K}$ has the desired properties over K if $\#(K) \geq \kappa_i$. Set $\kappa = \max\{\kappa_i\}$. Then if the cardinality of K is larger than κ we see that each $U_{i,K}$ satisfies the conclusions of the lemma. Hence X_K is Jacobson by Properties, Lemma 01P4. The statement on residue fields at closed points of X_K follows from the corresponding statements for residue fields of closed points of the $U_{i,K}$. □